

Ziyue Zeng

Ph.D. Student, Waseda University

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Research Interests

Generative compression · Compressed representation understanding ·
Video generation and frame interpolation · Deepfake detection

Education

Ph.D. in Fundamental Science and Engineering Waseda University, Kato Laboratory	Sep 2026 – Present Tokyo, Japan
M.S. in Fundamental Science and Engineering Waseda University, Watanabe Laboratory	Sep 2024 – Jul 2026 Tokyo, Japan
B.S. in Artificial Intelligence Chongqing University; exchange student at Waseda University, Apr – Sep 2023	Sep 2020 – Jul 2024 Chongqing, China

Research Experience

Research Intern, CG Lab Huawei Tokyo Da Vinci Research Institute	Jun 2026 – Present Tokyo, Japan
Research Assistant National Institute of Information and Communications Technology (NICT)	Apr 2025 – Apr 2026 Tokyo, Japan
Diffusion-based keyframe interpolation for low-bitrate video coding: Bi-AGMI (IEEE GCCE 2025, Oral Presentation Award) and FRS (IEVC 2026).	
Graduate Researcher, Watanabe Laboratory Waseda University	Sep 2024 – Jul 2026 Tokyo, Japan
GVCC: zero-shot video compression with a pretrained video generative model as the decoder (NeurIPS 2026); GVCCTurbo schedules its rate-compute trade-off. TSG: a universal synthesized image detector built on pretrained diffusion features (ACM MMAsia 2025, Oral). - Co-authored five papers on scalable image and video coding for humans and machines (IEEE MMSP, GCCE, ICCE; MIRU).	
Research Assistant, Information Processing Lab Chongqing University	Sep 2021 – Jun 2024 Chongqing, China
Evidence theory and information fusion; two first-author journal articles (<i>Chaos, Solitons & Fractals</i>; <i>Computational and Applied Mathematics</i>).	

Selected Publications

- [1] **Ziyue Zeng**, X. Su, H. Liu, B. Lu, Y. Tatsumi, H. Watanabe. **GVCC: Zero-Shot Video Compression via Codebook-Driven Stochastic Rectified Flow**. *NeurIPS 2026*, to appear.
- [2] **Ziyue Zeng**, D. Peng, X. Su, H. Watanabe. **GVCCTurbo: Rate-Compute Quality Scheduling for Codebook Driven Generative Compression**. *arXiv:2608.03517*, 2026.
- [3] **Ziyue Zeng**, Y. Tatsumi, H. Watanabe. **Flow Residual Segmentation and Generative Reconstruction for Motion-Aware Video Coding**. *IEVC 2026*.
- [4] **Ziyue Zeng**, Y. Guo, H. Liu, D. Peng, H. Watanabe. **Time Step Generating: A Universal Synthesized Deepfake Image Detector**. *ACM MMAsia 2025*, **Oral**.
- [5] **Ziyue Zeng**, Y. Tatsumi, H. Watanabe. **Bidirectional Attention-Gated Motion Injection for Frame Interpolation**. *IEEE GCCE 2025*, **Oral Presentation Award**.

Full publication list: ccdydy.github.io

Awards

IEEE GCCE 2025 Oral Presentation Award (Bi-AGMI, first author)	2025
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Skills

Python, PyTorch, Diffusers · Fine-tuning and inference of video diffusion and rectified-flow models (SVD, Wan) ·
LaTeX, scientific plotting